

Total No. of Questions : 4]

SEAT No. :

PD-265

[Total No. of Pages : 2

[6411]-40

**B.E. (Computer Engineering) (Insem.)  
HIGH PERFORMANCE COMPUTING  
(2019 Pattern) (Semester - VIII) (410250)**

**Time : 1 Hour]**

**[Max. Marks : 30**

**Instructions to the candidates :**

- 1) Answer Q. 1 or Q. 2, Q. 3 or Q. 4
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** What are types of dataflow execution models? [6]
- b) Explain SIMD, MIMD and SIMT architectures. [5]
- c) Write a short note on Level of Parallelism. [4]

OR

- Q2) a)** Explain N-wide Superscalar Architectures. [6]
- b) Explain Cache coherence in multiprocessor system. [5]
- c) What are the application of parallel computing? [4]

- Q3) a)** Explain any three-decomposition technique with example. [6]
- b) Explain classification of Dynamic mapping techniques. [5]
- c) What are characteristics of task and interaction? [4]

OR

P.T.O.

- Q4) a)** Explain with example (any two): **[6]**
- i) Recursive decomposition
  - ii) Data decomposition
  - iii) Exploratory decomposition
- b) Explain different schemes for Static Mapping. **[5]**
- c) Give the characteristics of task. **[4]**

